

Lab Book

Bradley Day

2025-05-19

Table of contents

Preface	3
I Summer Reading	4
1 Work Done Over the Summer Break	5
II Teaching Block 1	6
2 Week 1	7
2.1 2026/09/21	7
2.2 2026/09/22	7
2.2.1 Meeting Minutes	7
2.3 2026/09/23	8
2.3.1 Reading - K. Nandra (2007) [1]	8
III Teaching Block 2	20
References	31

Preface

This book details my research during the final year of my degree at the University of Bristol. Thank you to Professor Andrew Young for supervising and my project partner Jacob Taylor for his work.

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

Part I

Summer Reading

1 Work Done Over the Summer Break

Some reading and meetings occurred over the summer break prior to the start of the project, here I detail notes from meetings and will note key points from reading.

This project will be based on emission line fitting to the Iron $K\alpha$ emission line seen when looking at black holes. The focus will be on Active Galactic Nuclei (AGN).

Part II

Teaching Block 1

2 Week 1

2.1 2026/09/21

Lab book set up and initialised as a Github page. Meeting with supervisor arranged for 2026/09/22.

2.2 2026/09/22

Plans/questions for meeting:

- Confirm that Quarto is allowed for writing lab book
 - Yes
- Get connected to solva
 - This will need to be after Rhys has set things up
 - Note 2026-09-23 have connected to solva, however will wait until Rhys has given an induction to use
- Make sure GitHub set up properly so we can commit/push to it
- Confirm scope of project and form initial plan with Jacob
 - Fitting models?
 - * XSPEC?
 - * Julia (SpectralFitting.jl)?
 - * Other?
 - Data Reduction
 - * Using XMM-Newton data so will need to become familiar with [XMM SAS](#)
 - * Are we using this on solva?
 - * Note to self: see project plan on Github for more resources

2.2.1 Meeting Minutes

- XMM is not a survey telescope so even random selections will usually be interesting objects
- Andy will add a link to Nandra's paper on the Github repo for bright sources
 - Read this paper
- Browse paper(s) to find interesting sources
 - NGC4151?
 - Make sure to note when and how long observations were taken
 - HEASARC archive should let us search for particular objects
 - Download data and go through the ABC guide
- Before next meeting:
 - Look into candiadate AGN

- * Look at Tartarus database?
- Read papers provided by Andy
- Look through XMM SAS ABC guide

2.3 2026/09/23

2.3.1 Reading - K. Nandra (2007) [1]

- Covers 37 observations of 26 Seyfert galaxies from XMM-Newton
 - All show Iron K emission, which makes this paper useful for candidate selection for our research
 - Around 45% of the data was best fit with a relativistically blurred reflection model
 - * Some objects may not show the broadening?
- General background information
 - Iron line emission broadening predicted in Fabian et al (1989)
 - This was alongside the discovery that the emission likely comes from the accretion disk
 - Useful reads?
 - * Fabian (1989)
 - * George, Nandra & Fabian (1990)
 - * Tanaka et al. (1995)
- Data Reduction
 - They give a lot of detail on their data reduction which may be useful further down the line
 - Source taken from a circle with radius 700 detector pixels
 - There are several windowing modes on the CCD which will be important to note
- Model
 - They found that the minimum set of models was:
 - * Power law continuum (modelling the corona)
 - * Reflection model (e.g. XILLVER)
 - * Can use relativistic blurring on this however not necessarily needed
 - * Ionized absorber model
 - The models used are covered in depth, including how they selected if and how to apply the absorber models
 - * This will be useful later down the line to give some direction when fitting
 - They modelled the emission using a broad and narrow component, this may be useful to consider?
- Potential candidate list (from Figure 3 of Nandra et al.)
 - NGC3516(2)
 - * Very broad emission centred around the Fe K α line
 - NGC3783(2)
 - * Quite a clean spectrum with low noise/small errors
 - NGC4151(2) or NGC4151(3)
 - * Clean spectrum with broad emission

Part III

Teaching Block 2

References

- [1] K. Nandra *et al.*, Monthly Notices of the Royal Astronomical Society **382**, 194 (2007).